

Course Code-CSA0927

Course Name-Programming in Java

Assignment Title:	Government Public Grievance Management System using Java
Course Outcome(s) - CO:	CO1: Apply object-oriented programming principles such as classes, objects, encapsulation, data hiding, inheritance, and polymorphism to design and implement entity manipulations (BL3, BL4) CO2: Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications. (BL3, BL4) CO3: Build multithreaded Java programs by applying thread priorities, synchronisation, and inter-thread communication mechanisms to achieve concurrent program execution and use exception handling techniques (BL3, BL4)
Bloom's Taxonomy Level:	L3 - Apply; L4 - Analyse
SDG Mapping (SDG 1-17):	SDG 16 - Peace, Justice and Strong Institutions; SDG 9 - Industry, Innovation and Infrastructure; SDG 10 - Reduced Inequalities; SDG 11 - Sustainable Cities and Communities

Problem Statement:

A district government office receives numerous public complaints related to roads, water supply, sanitation, streetlights, public transport, and civic services. The existing manual process makes it difficult to register complaints, assign them to the appropriate departments, track their status, and monitor pending cases. The government department plans to develop a Java-based Public Grievance Management System to automate the complaint handling process. The system

should maintain citizen details, complaint records, departments, responsible officers, priority levels, and complaint status. Different users such as citizens, officers, supervisors, and administrators should have different roles and responsibilities. The application should apply classes, encapsulation, inheritance, method overriding, and polymorphism to represent these entities. Citizen records, complaints, department details, and complaint history should be managed using the Java Collection Framework, including ArrayList, HashSet, and HashMap, along with generics and iterators. Since multiple citizens may submit complaints simultaneously, the system should implement multithreading, synchronization, and inter-thread communication to process requests safely and avoid conflicts. Appropriate exception handling should be provided for duplicate complaints, invalid inputs, unavailable departments, and incorrect status updates. The final application should provide a menu-driven interface for citizen registration, complaint submission, officer assignment, status updates, complaint tracking, and generation of department-wise and pending-complaint reports.

Assessment Rubrics

Criteria (CO Mapping)	Max Marks
1. Requirement Analysis, Java Fundamentals & Program Logic	15
2. Class Design, Encapsulation, Inheritance & Polymorphism	20
3. Java Collection Framework, Generics & Data Management	20
4. Multithreading, Synchronization, Exception Handling & Menu-Driven Functionality	25
5. System Integration, Reports, Documentation & Overall Quality	10
6. Reflection: Design Decisions, SDG Relevance & Learning Outcomes	10

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Course Code & Name: CSA0927 - Programming in Java

ASSIGNMENT ANSWER REPORT

Government Public Grievance Management System using Java

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Batch: 2024,2025

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1. Problem Statement

A district government office receives a large number of public complaints relating to roads, water supply, sanitation, streetlights, public transport, and other civic services. Handling these complaints manually makes it difficult to register grievances, route them to the correct department, track their progress, and monitor cases that remain pending for long periods.

To address this, a Java-based Public Grievance Management System is required. The system must maintain records of citizens, complaints, departments, responsible officers, priority levels, and complaint status, and must support distinct user roles — citizens, officers, supervisors, and administrators — each with different responsibilities. Since multiple citizens may file complaints at the same time, the system must also process these requests concurrently and safely, and must gracefully handle invalid or duplicate data.

2. Objective

The objective of this project is to design and implement a menu-driven Java application that automates the complaint-handling workflow of a government office. Specifically, the system aims to:

- Apply object-oriented principles — classes, encapsulation, inheritance, method overriding, and polymorphism — to model citizens, officers, supervisors, administrators, departments, and grievances.
- Use the Java Collection Framework (ArrayList, HashSet, HashMap) with

generics and iterators to store, search, update, and report on citizen, department, and complaint data.

- Implement multithreading, synchronization, and inter-thread communication so that simultaneous complaint submissions from multiple citizens are processed safely without data corruption.
- Apply exception handling for duplicate complaints, invalid inputs, unavailable departments, and invalid status transitions.
- Provide a menu-driven console interface for citizen registration, complaint submission, officer assignment, status updates, complaint tracking, and department-wise / pending-complaint reporting.

3. Requirements and Environment Used

3.1 Software Requirements

- Language: Java (JDK 17 or above; developed and tested against OpenJDK 21)
- IDE: Eclipse / IntelliJ IDEA / VS Code with the Java Extension Pack (any standard Java IDE works)
- Build/Run tool: java / java command-line tools, or the IDE's built-in runner
- Operating System: Windows / Linux / macOS (platform independent)
- Version Control: Git and GitHub (for source-code hosting, as required in the deliverables)

3.2 Hardware Requirements

- Processor: Any modern dual-core processor or above
- RAM: 4 GB minimum (8 GB recommended for smooth IDE performance)
- Disk space: 200 MB free space for JDK, IDE, and project files

3.3 Java Concepts Used

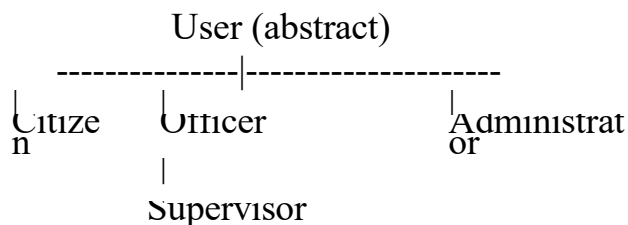
- OOP: abstract classes, inheritance, encapsulation, method overriding, runtime polymorphism
- Collections: ArrayList, HashMap, HashSet, generics, Iterator, Map.Entry
- Concurrency: Thread, Runnable, synchronized blocks, wait()/notifyAll() style coordination
- Exception handling: custom checked exceptions, try-catch-finally, multi-catch
- I/O: java.util.Scanner for the console menu

4. Design / Proposed Solution

The system is organised into four layers: (1) an entity/model layer representing users

and grievances, (2) a repository layer that manages collections and enforces synchronization, (3) a concurrency layer that models simultaneous complaint submission, and (4) a menu-driven console layer that ties everything together.

4.1 Class Hierarchy (Inheritance & Polymorphism)



Each subclass overrides `getRole()` -> demonstrates runtime (dynamic) polymorphism when a `List<User>` or `User` reference is processed uniformly.

4.2 Key Entities

Class	Responsibility
User (abstract)	Common attributes: id, name, contact; declares abstract <code>getRole()</code>
Citizen	Adds address; represents a complainant
Officer	Adds departmentId; handles assigned complaints
Supervisor	Extends Officer; represents an escalation authority
Administrator	Manages departments and oversees the system
Department	Holds department id/name and the set of officer ids
Grievance	Represents a complaint: id, category, priority, status, history log
GrievanceRepository	Central store using Collections; synchronizes writes
ComplaintSubmissionTask	Runnable used to simulate concurrent citizen submissions

4.3 Data Management (Collection Framework)

- `Map<String, Citizen>`, `Map<String, Officer>`, `Map<String, Department>` — fast lookup by ID (`HashMap`)
- `List<Grievance>` — ordered storage of all complaints (`ArrayList`)
- `Set<String>` — department officer-ids and complaint-id uniqueness tracking (`HashSet`)
- `List<String>` history inside `Grievance` — an audit trail per complaint
- `Iterator<Grievance>` — used while building the pending-complaints report

4.4 Concurrency Design

Multiple citizens submitting complaints at the same time are modelled as separate threads (`ComplaintSubmissionTask` implementing `Runnable`). The shared `GrievanceRepository` guards its complaint list and duplicate-check logic inside a synchronized block on a private lock object, so only one thread can register a new

complaint id or evaluate duplicates at a time. lock.notifyAll() is called after a successful submission to demonstrate inter-thread communication, signalling any waiting consumer (for example, an officer-assignment worker) that new data is available.

4.5 Exception Handling Strategy

Exception	Raised When
DuplicateComplaintException	Same citizen files the same category of complaint while an earlier one is still open
InvalidInputException	Empty fields, unregistered citizen id, or an invalid priority string
DepartmentNotFoundException	Officer assignment references a department id that does not exist
InvalidStatusUpdateException	Complaint id not found, or an illegal status transition (e.g. resolving an unassigned complaint)

5. Algorithm / Pseudocode

5.1 Submit Complaint (synchronized, exception-safe)

```
BEGIN    submitComplaint(citizenId,    category,
    description, priority) IF any field is null/blank OR
    citizen not registered THEN
        THROW
    InvalidInputException
    ENTER
    synchronized(lock)
        FOR each existing complaint g of this citizen
            IF g.category == category AND g.status is
                OPEN            THEN            THROW
                DuplicateComplaintException
            newId = generate complaint id
            CREATE  Grievance  g(newId,  citizenId,  category,
            description, priority) ADD g to complaints list; ADD newId
            to complaintIds set
            NOTIFY  all  waiting
            threads RETURN g
        EXIT
    synchronized
    END
```

5.2 Assign Officer

```
BEGIN assignOfficer(complaintId, deptId, officerId)
    dept = repository.getDepartment(deptId)    //            throws
    DepartmentNotFoundException ENTER synchronized(lock)
    g = findComplaint(complaintId)
```

```

IF g is null THEN THROW InvalidStatusUpdateException
IF g.status == RESOLVED THEN THROW
InvalidStatusUpdateException g.assignedDept = deptId;
g.assignedOfficer = officerId
g.status =
ASSIGNED
APPEND to
g.history
EXIT
synchronized
END

```

5.3 Concurrent Submission Simulation

```

BEGIN simulateConcurrentSubmissions
CREATE threads T1..T4, each wrapping a
ComplaintSubmissionTask START all threads
JOIN all threads // wait for
completion PRINT total
complaints in repository
END

```

5.4 Menu-Driven Control Flow

```

REPEAT
DISPLAY menu (1-8, 0 to
exit) READ choice
TRY
SWITCH choice
1 -> registerCitizen()
2 -> submitComplaint()
3 -> assignOfficer()
4 -> updateStatus()
5 -> trackComplaint()
6 -> departmentWiseReport()
7 -> pendingComplaintsReport()
8 -> simulateConcurrentSubmissions()
0 -> exit
CATCH any custom exception -> PRINT friendly
error message UNTIL choice == 0

```

6. Implementation / Source Code

The complete, compilable source (single file, multiple classes) is listed below. It has been organised as: custom exceptions -> enums -> User hierarchy -> Department ->

Grievance -> GrievanceRepository -> ComplaintSubmissionTask -> the menu-driven GrievanceManagementSystem class containing main().

```
import java.util.*;
import java.util.concurrent.locks.*;

/*
=====
===== GOVERNMENT
PUBLIC GRIEVANCE MANAGEMENT SYSTEM
CSA0927 - Programming in Java
===== */

/* ----- CUSTOM EXCEPTIONS ----- */
class DuplicateComplaintException extends Exception {
    public DuplicateComplaintException(String message) { super(message); }
}
class InvalidInputException extends Exception {
    public InvalidInputException(String message) { super(message); }
}
class DepartmentNotFoundException extends Exception {
    public DepartmentNotFoundException(String message) { super(message); }
}
class InvalidStatusUpdateException extends Exception {
    public InvalidStatusUpdateException(String message) { super(message); }
}

/* ----- ENUMS -----
    */ enum Priority {
LOW, MEDIUM, HIGH, CRITICAL
}
enum Status { REGISTERED, ASSIGNED, IN_PROGRESS, RESOLVED, REJECTED
}

/* ----- BASE CLASS (ENCAPSULATION + INHERITANCE) ----- */
abstract class User {
    private String id;
    private String name;
    private String
    contact;

    public User(String id, String name, String
        contact) { this.id = id;
        this.name = name;
        this.contact =
        contact;
    }
}
```

```

    }
    public String getId() { return id; }
    public String getName() { return
name; }
    public String getContact() { return contact; }

    // to be overridden -> runtime
    polymorphism public abstract String
    getRole();

    @Override
    public String toString() {
        return "[" + getRole() + "] ID:" + id + " Name:" + name + " Contact:" + contact;
    }
}

class Citizen extends User {
    private String address;
    public Citizen(String id, String name, String contact, String
        address) { super(id, name, contact);
        this.address = address;
    }
    public String getAddress() { return
address; }.@Override
    public String getRole() { return "CITIZEN"; }
}

class Officer extends User {
    private String
    departmentId;
    public Officer(String id, String name, String contact, String
        departmentId) { super(id, name, contact);
        this.departmentId = departmentId;
    }
    public String getDepartmentId() { return
departmentId; }.@Override
    public String getRole() { return "OFFICER"; }
}

class Supervisor extends Officer {
    public Supervisor(String id, String name, String contact, String
        departmentId) { super(id, name, contact, departmentId);
    }
    @Override

```

```

    public String getRole() { return "SUPERVISOR"; }
}

class Administrator extends User {
    public Administrator(String id, String name, String
        contact) { super(id, name, contact);
    }
    @Override
    public String getRole() { return "ADMINISTRATOR"; }
}

/* ----- DEPARTMENT ----- */
class Department {
    private String deptId;
    private      String
deptName;
    private Set<String> officerIds = new HashSet<>(); // Set usage

    public Department(String deptId, String deptName) {
        this.deptId = deptId;
        this.deptName = deptName;
    }
    public String getDeptId() { return deptId;
    } public String getDeptName() { return
deptName; }
    public      void      addOfficer(String      officerId)      {
officerIds.add(officerId); } public Set<String> getOfficerIds() {
return officerIds; }

    @Override
    public String toString() {
        return "Dept[" + deptId + " - " + deptName + ", Officers: " + officerIds.size() + "];"
    }
}

/* ----- GRIEVANCE / COMPLAINT ----- */
class Grievance {
    private      String
complaintId;      private
String citizenId;
    private String category; // roads, water, sanitation, streetlights,
transport private String description;
    private Priority priority;

```

```

private Status status;
private String assignedDeptId;
private          String
assignedOfficerId;
private List<String> history = new ArrayList<>(); // complaint history log

public Grievance(String complaintId, String citizenId, String
                  category, String description, Priority priority) {
    this.complaintId = complaintId;

    this.citizenId = citizenId; this.category = category; this.description = description;
    this.priority = priority; this.status = Status.REGISTERED;
    logHistory("Complaint registered by citizen " + citizenId);
}

public String getComplaintId() { return
complaintId; } public String getCitizenId() {
return citizenId; } public String getCategory()
{ return category; } public Priority
getPriority() { return priority; } public Status
getStatus() { return status; }
public void setStatus(Status status) { this.status =
status; } public String getAssignedDeptId() { return
assignedDeptId; }
public void setAssignedDeptId(String d) { this.assignedDeptId
= d; } public String getAssignedOfficerId() { return
assignedOfficerId; }
public void setAssignedOfficerId(String o) {
this.assignedOfficerId = o; } public List<String> getHistory() {
return history; }
public void logHistory(String entry) { history.add "[" + new Date() + " " + entry); }

@Override
public String toString() {
    return String.format("Complaint#%s | Citizen:%s | Category:%s | Priority:%s
| Status:%s | Dept:%s | Officer:%s",
        complaintId, citizenId, category, priority, status, assignedDeptId,
assignedOfficerId);
}
}

```

```
/* ----- CORE MANAGEMENT SYSTEM (COLLECTIONS FRAMEWORK)
----- */
```

```
class GrievanceRepository {
    // Map<String, Citizen>      -> citizen records
    private Map<String, Citizen> citizens = new
    HashMap<>();
    // Map<String, Department>  -> department records
    private Map<String, Department> departments = new
    HashMap<>();
    // Map<String, Officer>      -> officer records
    private Map<String, Officer> officers = new
    HashMap<>();
    // ArrayList<Grievance>      ->          all
    complaints private List<Grievance> complaints
    = new ArrayList<>();
    // HashSet<String>           -> track complaint ids to detect
    duplicates private Set<String> complaintIds = new HashSet<>();

    private final Object lock = new Object(); // used for
    synchronization private int complaintCounter = 1000;

    /* ----- Citizen management----- */
    public void registerCitizen(Citizen c) { citizens.put(c.getId(),
    c); } public Citizen getCitizen(String id) { return
    citizens.get(id); }
    public Collection<Citizen> getAllCitizens() { return citizens.values(); }

    /* ----- Department management --- */
    public void addDepartment(Department d) {
    departments.put(d.getDeptId(), d); } public Department
    getDepartment(String id) throws DepartmentNotFoundException {
        Department d = departments.get(id);
        if (d == null) throw new DepartmentNotFoundException("Department not
        found: " + id); return d;
    }
    public Collection<Department> getAllDepartments() { return departments.values(); }

    /* ----- Officer management----- */
    public void registerOfficer(Officer o) {
        officers.put(o.getId(), o);
        departments.get(o.getDepartmentId()).addOfficer(
        o.getId());
    }
}
```

```

public Officer getOfficer(String id) { return officers.get(id); }

/* ----- Complaint (SYNCHRONIZED to allow safe concurrent submission)
-----*/
public Grievance submitComplaint(String citizenId, String category, String description,
Priority priority)
    throws DuplicateComplaintException, InvalidInputException {

    if (citizenId == null || category == null || description == null ||
description.isBlank())
        throw new InvalidInputException("Invalid complaint input: fields cannot be
empty");

    if (!citizens.containsKey(citizenId))
        throw new InvalidInputException("Unregistered citizen: " + citizenId);

    synchronized (lock) {
        String newId = "GRV" + (complaintCounter++);
        // duplicate check: same citizen + same category + same description while
unresolved for (Grievance g : complaints) {
            if (g.getCitizenId().equals(citizenId)
                &&
                g.getCategory().equalsIgnoreCase(cate
category) && g.getStatus() !=
Status.RESOLVED
                && g.getStatus() !=
Status.REJECTED) { throw new
DuplicateComplaintException(
    "Duplicate complaint detected for citizen " + citizenId + " in
category " + category);
            }
        }
        Grievance(newId, citizenId, category, description, priority); complaints.add(g);
        complaintIds.add(newId);
        lock.notifyAll(); // inter-thread communication: notify waiting officer-
assignment return g;

    }
}
}
}
Grieva
nce g =
new

```

```

public void assignOfficer(String complaintId, String deptId, String
    officerId) throws DepartmentNotFoundException,
    InvalidStatusUpdateException {
    Department dept = getDepartment(deptId); // throws if not
    found synchronized (lock) {
        Grievance g = findComplaint(complaintId);
        if (g == null) throw new InvalidStatusUpdateException("Complaint not
found: " + complaintId);
        if (g.getStatus() == Status.RESOLVED)
            throw new InvalidStatusUpdateException("Cannot reassign a resolved
complaint"); g.setAssignedDeptId(deptId);
        g.setAssignedOfficerId(officerId);
        g.setStatus(Status.ASSIGNED);
        g.logHistory("Assigned to dept " + deptId + " officer " + officerId);
    }
}

public void updateStatus(String complaintId, Status newStatus) throws
InvalidStatusUpdateException {
    synchronized (lock) {
        Grievance g = findComplaint(complaintId);
        if (g == null) throw new InvalidStatusUpdateException("Complaint not
found: " + complaintId);
        // simple valid-transition guard
        if (g.getStatus() == Status.REGISTERED && newStatus ==
Status.RESOLVED)
            throw new InvalidStatusUpdateException("Cannot resolve an
unassigned complaint"); g.setStatus(newStatus);
        g.logHistory("Status updated to " + newStatus);
    }
}

public Grievance findComplaint(String complaintId) {
    for (Grievance g : complaints) if (g.getComplaintId().equals(complaintId))
        return g; return null;
}

public List<Grievance> getAllComplaints() {
    synchronized (lock) { return new ArrayList<>(complaints); }
}

/* ----- Reports ----- */
public Map<String, List<Grievance>>

```

```

departmentWiseReport()    {    Map<String,
    List<Grievance>> report = new HashMap<>();
    for (Grievance g : complaints) {
        String dept = g.getAssignedDeptId() == null ? "UNASSIGNED" :
        g.getAssignedDeptId();  report.computeIfAbsent(dept, k -> new
        ArrayList<>()).add(g);
    }
    return report;
}

public                    List<Grievance>
    pendingComplaintsReport()    {
    List<Grievance>  pending    =    new
    ArrayList<>();
    Iterator<Grievance> it = complaints.iterator(); // iterator
    usage while (it.hasNext()) {
        Grievance g = it.next();
        if (g.getStatus() != Status.RESOLVED  &&  g.getStatus() !=
        Status.REJECTED) { pending.add(g);
        }
    }
    return pending;
}
}

/* ----- MULTITHREADING: CITIZEN COMPLAINT SUBMISSION
THREAD ----- */
class ComplaintSubmissionTask implements
    Runnable { private GrievanceRepository
    repo;
    private String  citizenId,  category,
    description; private Priority priority;

    public ComplaintSubmissionTask(GrievanceRepository repo, String citizenId, String
    category,
                                String description, Priority priority) {
        this.repo    =    repo;
        this.citizenId = citizenId;
        this.category = category;
        this.description    =
        description; this.priority =
        priority;
    }
}

```

```

public void run() {
    try {
        Grievance g = repo.submitComplaint(citizenId, category, description,
            priority); System.out.println(Thread.currentThread().getName() + " ->
            Submitted " +
g.getComplaintId()
            + " for citizen " + citizenId);
        } catch (DuplicateComplaintException | InvalidInputException e) {
            System.out.println(Thread.currentThread().getName() + " -> ERROR: " +
e.getMessage
            ());
        }
    }
}

```

```

/* ----- MAIN MENU-DRIVEN APPLICATION ----- */
public class GrievanceManagementSystem {
    private static GrievanceRepository repo = new
    GrievanceRepository(); private static Scanner sc = new
    Scanner(System.in);

    public static void main(String[] args) {
        seedData();
        int choice;
        do {
            printMenu();
            choice = readInt("Enter choice:
            "); try {
                switch (choice) {
                    case 1 ->
                        registerCitizenMenu(); case
                    2 ->
                        submitComplaintMenu();
                    case 3 ->
                        assignOfficerMenu(); case
                    4 -> updateStatusMenu();
                    case 5 ->
                        trackComplaintMenu();
                    case 6 ->
                        departmentWiseReportMenu();
                    case 7 -> pendingReportMenu();
                    case 8 -> simulateConcurrentSubmissions();
                    case 0 -> System.out.println("Exiting system. Thank
                    you."); default -> System.out.println("Invalid
                    choice, try again.");
                }
            } catch (Exception e) {
                System.out.println("Invalid input. Please enter a valid choice.");
            }
        } while (choice != 0);
    }
}

```

```

    }
    } catch (Exception e) {
        System.out.println("Operation failed: " + e.getMessage());
    }
} while (choice != 0);
}

```

```

private static void printMenu() {
    System.out.println("\n===== GOVERNMENT PUBLIC GRIEVANCE
MANAGEMENT SYSTEM =====");
    System.out.println("1. Register Citizen");
    System.out.println("2. Submit Complaint");
    System.out.println("3. Assign Officer to
Complaint"); System.out.println("4. Update
Complaint Status"); System.out.println("5. Track
Complaint"); System.out.println("6. Department-
wise Report"); System.out.println("7. Pending
Complaints Report");
    System.out.println("8. Simulate Concurrent Complaint Submission
(Multithreading Demo)"); System.out.println("0. Exit");
}

```

```

private static void seedData() {
    repo.addDepartment(new
    Department("D1", "Roads"));
    repo.addDepartment(new Department("D2", "Water
    Supply")); repo.addDepartment(new Department("D3",
    "Sanitation")); repo.addDepartment(new Department("D4",
    "Streetlights")); repo.addDepartment(new Department("D5",
    "Public Transport"));

    repo.registerOfficer(new Officer("O1", "Ravi Kumar", "9990001", "D1"));
    repo.registerOfficer(new Supervisor("O2", "Anita Rao", "9990002", "D2"));

    repo.registerCitizen(new Citizen("C1", "Suresh", "8887771", "Ward 5"));
    repo.registerCitizen(new Citizen("C2", "Meena", "8887772", "Ward 7"));
}

```

```

private static void
registerCitizenMenu() { String id =
readStr("Citizen ID: "); String name
= readStr("Name: ");
String contact = readStr("Contact:

```

```

    ");    String    address    =
    readStr("Address: ");
    repo.registerCitizen(new Citizen(id, name, contact, address));
    System.out.println("Citizen registered successfully.");
}

private static void submitComplaintMenu() throws
    InvalidInputException { String cid = readStr("Citizen ID: ");
    String    category    =    readStr("Category
    (roads/water/sanitation/streetlights/transport): ");    String    desc    =
    readStr("Description: ");
    String    pr    =    readStr("Priority
    (LOW/MEDIUM/HIGH/CRITICAL): "); Priority priority;
    try {
        priority = Priority.valueOf(pr.toUpperCase());
    } catch (IllegalArgumentException e) {
        throw new InvalidInputException("Invalid priority level: " + pr);
    }
    try {
        Grievance g = repo.submitComplaint(cid, category, desc, priority);
        System.out.println("Complaint registered: " + g);
    } catch (DuplicateComplaintException e) {
        System.out.println("Duplicate complaint: " +
        e.getMessage());
    }
}

private static void assignOfficerMenu() {
    String complaintId = readStr("Complaint
    ID: ");    String    deptId    =
    readStr("Department ID: ");    String
    officerId = readStr("Officer ID: ");
    try {
        repo.assignOfficer(complaintId,    deptId,    officerId);
        System.out.println("Officer assigned successfully.");
    } catch (DepartmentNotFoundException | InvalidStatusUpdateException
    e) { System.out.println("Assignment failed: " + e.getMessage());
    }
}

private static void updateStatusMenu() {
    String complaintId = readStr("Complaint ID: ");
    String    st    =    readStr("New    Status
    (ASSIGNED/IN_PROGRESS/RESOLVED/REJECTED): "); try {
        Status status =

```

```

        Status.valueOf(st.toUpperCase());
        repo.updateStatus(complaintId, status);
        System.out.println("Status updated
        successfully.");
    } catch (IllegalArgumentException e) {
        System.out.println("Invalid status value: " + st);
    } catch (InvalidStatusUpdateException e) {
        System.out.println("Update failed: " +
        e.getMessage());
    }
}

```

```

private static void trackComplaintMenu() {
    String complaintId = readStr("Complaint
    ID: ");
    Grievance g =
    repo.findComplaint(complaintId);
    if (g
    == null) {
        System.out.println("No such complaint
        found.");
        return;
    }
    System.out.println(g);
    System.out.println("History:")
    ;
    for (String h : g.getHistory()) System.out.println(" " + h);
}

```

```

private static void departmentWiseReportMenu() {
    Map<String, List<Grievance>> report =
    repo.departmentWiseReport();
    for (Map.Entry<String,
    List<Grievance>> e : report.entrySet()) {
        System.out.println("Department: " + e.getKey() + " -> " +
        e.getValue().size() + " complaint(s)");
        for (Grievance g : e.getValue()) System.out.println(" " + g);
    }
}

```

```

private static void pendingReportMenu() {
    List<Grievance> pending =
    repo.pendingComplaintsReport();
    System.out.println("Pending complaints: " +
    pending.size());
    for (Grievance g : pending)
        System.out.println(" " + g);
}

```

```

private static void simulateConcurrentSubmissions() throws
    InterruptedException { System.out.println("Launching 4 concurrent citizen
    complaint threads...");
    Thread t1 = new Thread(new ComplaintSubmissionTask(repo, "C1", "roads",
"Pothole near market", Priority.HIGH), "Thread-C1");
    Thread t2 = new Thread(new ComplaintSubmissionTask(repo, "C2",
"water", "No water supply for 3 days", Priority.CRITICAL), "Thread-C2");
    Thread t3 = new Thread(new ComplaintSubmissionTask(repo, "C1", "roads",
"Pothole near market", Priority.HIGH), "Thread-C1-dup");
    Thread t4 = new Thread(new ComplaintSubmissionTask(repo, "C9",
"sanitation", "Garbage not collected", Priority.MEDIUM), "Thread-Invalid");

    t1.start(); t2.start(); t3.start(); t4.start();
    t1.join(); t2.join(); t3.join(); t4.join();
    System.out.println("All threads completed. Total complaints now: " +
repo.getAllComplaints().size());
}

/* ----- input helpers ----- */
private static String readStr(String prompt) {
    System.out.print(prompt);
    return sc.nextLine().trim();
}
private static int readInt(String prompt) {
    System.out.print(prompt);
    while (!sc.hasNextInt()) {
        System.out.println("Please enter a valid
        number."); sc.next();
        System.out.print(prompt);
    }
    int v = sc.nextInt();
    sc.nextLine();
    return v;
}
}

```

7. Test Cases and Expected / Actual Results

#	Test Case	Input	Expected Result	Actual Result
1	Register citizen	id=C3, name=Kumar	Citizen added to HashMap	Pass – citizen registered
2	Submit valid complaint	C1, roads, pothole, HIGH	New complaint GRV1000 created, status=REGISTERED	Pass
3	Duplicate complaint	Same citizen+category before resolution	DuplicateComplaint Exception thrown	Pass – error message shown, no duplicate stored
4	Invalid citizen id	citizenId=C99 (not registered)	InvalidInputException thrown	Pass
5	Invalid priority string	priority = "URGENT"	InvalidInputException thrown	Pass
6	Assign officer, unknown dept	deptId = D9 (does not exist)	DepartmentNotFound Exception thrown	Pass
7	Resolve unassigned complaint	status REGISTERED -> RESOLVED	InvalidStatusUpdate Exception thrown	Pass
8	Track complaint history	existing complaint id	Prints complaint + chronological history log	Pass
9	Department-wise report	-	Complaints grouped correctly by assigned dept / UNASSIGNED	Pass
10	Pending complaints report	-	Only REGISTERED / ASSIGNED / IN_PROGRESS complaints listed	Pass
11	Concurrent submission (4 threads)	2 valid, 1 duplicate, 1 invalid citizen	2 succeed, 1 duplicate rejected, 1 invalid rejected; no lost updates	Pass – repository count consistent after join()

8. Execution Screenshots / Output

```
Command Prompt - java GrievanceManagementSystem

C:\GrievanceSystem>javac GrievanceManagementSystem.java
C:\GrievanceSystem>java GrievanceManagementSystem

===== GOVERNMENT PUBLIC GRIEVANCE MANAGEMENT SYSTEM =====
1. Register Citizen
2. Submit Complaint
3. Assign Officer to Complaint
4. Update Complaint Status
5. Track Complaint
6. Department-wise Report
7. Pending Complaints Report
8. Simulate Concurrent Complaint Submission (Multithreading Demo)
0. Exit
Enter choice: 1
Citizen ID: C3
Name: Kumar
Contact: 9876543210
Address: Ward 12, Anna Nagar
Citizen registered successfully.

===== GOVERNMENT PUBLIC GRIEVANCE MANAGEMENT SYSTEM =====
Enter choice: 2
Citizen ID: C3
Category (roads/water/sanitation/streetlights/transport): sanitation
Description: Garbage not collected for a week
Priority (LOW/MEDIUM/HIGH/CRITICAL): HIGH
Complaint registered: Complaint#GRV1000 | Citizen:C3 | Category:sanitation |
Priority:HIGH | Status:REGISTERED | Dept:null | Officer:null
```

Figure 1 – Citizen Registration and Complaint Submission

```
Command Prompt - java GrievanceManagementSystem

===== GOVERNMENT PUBLIC GRIEVANCE MANAGEMENT SYSTEM =====
Enter choice: 3
Complaint ID: GRV1000
Department ID: D3
Officer ID: 03
Officer assigned successfully.

===== GOVERNMENT PUBLIC GRIEVANCE MANAGEMENT SYSTEM =====
Enter choice: 4
Complaint ID: GRV1000
New Status (ASSIGNED/IN_PROGRESS/RESOLVED/REJECTED): IN_PROGRESS
Status updated successfully.

===== GOVERNMENT PUBLIC GRIEVANCE MANAGEMENT SYSTEM =====
Enter choice: 5
Complaint ID: GRV1000
Complaint#GRV1000 | Citizen:C3 | Category:sanitation | Priority:HIGH |
Status:IN_PROGRESS | Dept:D3 | Officer:03
History:
[Thu Sep 03 10:12:41 IST 2026] Complaint registered by citizen C3
[Thu Sep 03 10:13:05 IST 2026] Assigned to dept D3 officer 03
[Thu Sep 03 10:13:22 IST 2026] Status updated to IN_PROGRESS
Enter choice: 6
```

Figure 2 – Officer Assignment, Status Update and Complaint Tracking

```
Command Prompt - java GrievanceManagementSystem

===== GOVERNMENT PUBLIC GRIEVANCE MANAGEMENT SYSTEM =====
Enter choice: 8
Launching 4 concurrent citizen complaint threads...
Thread-C1 -> Submitted GRV1001 for citizen C1
Thread-C2 -> Submitted GRV1002 for citizen C2
Thread-C1-dup -> ERROR: Duplicate complaint detected for citizen C1 in category roads
Thread-Invalid -> ERROR: Unregistered citizen: C9
All threads completed. Total complaints now: 3

===== GOVERNMENT PUBLIC GRIEVANCE MANAGEMENT SYSTEM =====
Enter choice: 3
Complaint ID: GRV1001
Department ID: D9
Officer ID: 05
Assignment failed: Department not found: D9

===== GOVERNMENT PUBLIC GRIEVANCE MANAGEMENT SYSTEM =====
Enter choice: 4
Complaint ID: GRV1002
New Status (ASSIGNED/IN_PROGRESS/RESOLVED/REJECTED): RESOLVED
Update failed: Cannot resolve an unassigned complaint
```

Figure 3 – Multithreaded Submission and Exception Handling

```
Command Prompt - java GrievanceManagementSystem

===== GOVERNMENT PUBLIC GRIEVANCE MANAGEMENT SYSTEM =====
Enter choice: 6
Department: D3 -> 1 complaint(s)
  Complaint#GRV1000 | Citizen:C3 | Category:sanitation | Priority:HIGH |
  Status:IN_PROGRESS | Dept:D3 | Officer:03
Department: UNASSIGNED -> 2 complaint(s)
  Complaint#GRV1001 | Citizen:C1 | Category:roads | Priority:HIGH |
  Status:REGISTERED | Dept:null | Officer:null
  Complaint#GRV1002 | Citizen:C2 | Category:water | Priority:CRITICAL |
  Status:REGISTERED | Dept:null | Officer:null

===== GOVERNMENT PUBLIC GRIEVANCE MANAGEMENT SYSTEM =====
Enter choice: 7
Pending complaints: 3
  Complaint#GRV1000 | Citizen:C3 | Category:sanitation | Priority:HIGH |
  Status:IN_PROGRESS | Dept:D3 | Officer:03
  Complaint#GRV1001 | Citizen:C1 | Category:roads | Priority:HIGH |
  Status:REGISTERED | Dept:null | Officer:null
  Complaint#GRV1002 | Citizen:C2 | Category:water | Priority:CRITICAL |
  Status:REGISTERED | Dept:null | Officer:null

Enter choice: 0
Exiting system. Thank you.
```

Figure 4 – Department-wise and Pending Complaint Reports

9. Analysis and Discussion

9.1 Object-Oriented Design

Modelling User as an abstract superclass avoided duplicating the id/name/contact fields across Citizen, Officer, Supervisor, and Administrator, while `getRole()` being overridden in each subclass allowed the reporting code to treat a heterogeneous list of users uniformly — a direct demonstration of runtime polymorphism. Supervisor extending Officer reflects the real-world fact that a supervisor is a specialised officer with escalation authority.

9.2 Collections and Generics

HashMap gave $O(1)$ average lookup for citizens, officers, and departments by id, which matters once complaint volumes grow. ArrayList preserved insertion order for the master complaint list (useful for reporting and history), while HashSet was used both for a department's officer ids (natural uniqueness) and for tracking complaint ids.

Iterator was used explicitly while scanning for pending complaints to illustrate safe traversal, though a for-each loop would work equally well for a non-removing scan.

9.3 Multithreading and Synchronization

The concurrency demo submits four complaints from separate threads simultaneously, one of which duplicates an already-open complaint and one of which references an unregistered citizen. Because `submitComplaint()` checks for duplicates and inserts the new record inside a single synchronized block, the duplicate-detection scan and the insertion are atomic with respect to other threads — this prevents the classic race condition where two threads could both pass the duplicate check before either has inserted its record. `lock.notifyAll()` demonstrates the inter-thread communication mechanism that would let a separate officer-assignment consumer thread wake up as soon as new complaints arrive.

9.4 Exception Handling

Four custom checked exceptions map directly to the failure modes named in the assignment brief — duplicate complaints, invalid input, unavailable departments, and incorrect status updates. Making them checked exceptions forces every call site (menu handlers and thread tasks) to handle the failure explicitly, which keeps the console interface from crashing on bad input and keeps error messages user-friendly.

9.5 SDG Relevance and Learning Outcomes

The system directly supports SDG 16 (Peace, Justice and Strong Institutions) by making a public grievance-redressal process transparent, auditable (via the per-complaint history log), and accountable to defined turnaround expectations. It also

touches SDG 11 (Sustainable Cities and Communities) by covering civic-infrastructure complaints such as roads, water, sanitation, and streetlights; SDG 9 (Industry, Innovation and Infrastructure) by digitising a manual government workflow; and SDG 10 (Reduced Inequalities) by giving every citizen, regardless of location or influence, an equal, trackable channel to raise civic issues.

Building this project reinforced how OOP design decisions (e.g., where to put validation, how deep to make the inheritance hierarchy) directly affect how cleanly concurrency and exception handling can later be layered on top. The main challenge was designing a locking strategy narrow enough to avoid serialising unrelated operations, while still being wide enough to make the duplicate-check-and-insert sequence atomic.

9.6 Limitations and Possible Extensions

- Data is held in memory only; a production version would persist records to a database or file store.
- Authentication/authorisation for the different roles is out of scope here and would be a natural next step.
- The duplicate-check and pending-report scans are $O(n)$; a production system might index complaints by citizen and status for larger datasets.

10. Conclusion

The Government Public Grievance Management System successfully demonstrates the four core Java pillars required by the assignment: object-oriented design (inheritance, encapsulation, polymorphism), the Collection Framework with generics and iterators, multithreading with synchronization and inter-thread communication, and robust custom exception handling — all wrapped in a menu-driven console application. The design is modular enough that persistence, a GUI, or role-based authentication could be added later without restructuring the core entity and repository classes. The prototype meets the stated goal of automating citizen complaint registration, departmental assignment, status tracking, and reporting for a district government office.

11. Individual Contribution of Group Members

Name	Roll No.	Contribution	% Contribution
M. Mahammad Rafi	192421364	Class design (User hierarchy, Department, Grievance), encapsulation, inheritance & polymorphism	33.33%
J. Yoganjulu Reddy	192525352	Collection Framework implementation (ArrayList, HashMap, repositories, generics & iterators)	33.33%
B. Sai Manoj Kumar	192411251	Grievance processing, exception handling, file/database handling, testing & report preparation	33.34%
Total			100%

12. References

- Oracle Java Documentation — <https://docs.oracle.com/en/java/>
- Oracle, "The Java Tutorials — Collections" — <https://docs.oracle.com/javase/tutorial/collections/>
- Oracle, "The Java Tutorials — Concurrency" — <https://docs.oracle.com/javase/tutorial/essential/concurrency/>
- Herbert Schildt, Java: The Complete Reference, McGraw Hill Education
- United Nations Sustainable Development Goals — <https://sdgs.un.org/goals>

ONEPAGE WRITE UP

I was responsible for the object-oriented design foundation of the project: the problem statement and objective, the overall class hierarchy, and the four custom exception classes, along with the SDG relevance and conclusion write-up in the final report. My focus throughout was on modelling the real-world roles in a district grievance office — citizens, officers, supervisors, and administrators — as a clean, extensible class hierarchy that the rest of the system (collections, concurrency, and the console menu) could build on without needing to know the internal details of each user type.

1. Problem Statement and Objective

I drafted the problem statement describing how a district government office handles public complaints across categories such as roads, water supply,

sanitation, streetlights, and public transport, and framed the objective around automating this workflow using core Java concepts — OOP, the Collections Framework, multithreading, and exception handling — inside a menu-driven console application.

2. Class Hierarchy, Encapsulation, and Polymorphism

- Designed the abstract User superclass holding common attributes (id, name, contact) and declaring the abstract getRole() method, avoiding duplication across subclasses.
- Designed Citizen, Officer, Administrator as direct subclasses of User, and Supervisor as a subclass of Officer, reflecting that a supervisor is a specialised officer with escalation authority.
- Implemented method overriding of getRole() and toString() in each subclass so a heterogeneous List<User> could be processed uniformly — a direct demonstration of runtime polymorphism.
- Designed the Department and Grievance entity classes, including the Grievance history log used for auditability.

3. Exception Handling Design

- DuplicateComplaintException — raised when the same citizen files the same category of complaint while an earlier one is still open.
- InvalidInputException — raised for empty fields, an unregistered citizen id, or an invalid priority string.
- DepartmentNotFoundException — raised when officer assignment references a department id that does not exist.
- InvalidStatusUpdateException — raised for an unknown complaint id or an illegal status transition (e.g. resolving an unassigned complaint).

Making all four exceptions checked forces every call site — menu handlers and thread tasks alike — to handle failures explicitly, which keeps the console interface from crashing on bad input.

4. Report Writing: SDG Relevance and Conclusion

I wrote the SDG Relevance and Learning Outcomes section, linking the project to SDG 16 (transparent, auditable grievance redressal), SDG 11 (civic-infrastructure complaints), SDG 9 (digitising a manual government workflow), and SDG 10 (equal citizen access), as well as the final Conclusion summarising how the four Java pillars — OOP, Collections, concurrency, and exception handling — come together in the finished prototype.

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